JAVA WITH DSA-DAY 5

HACKATHON

BANK MANAGEMENT SYSTEM:

```
import java.util.*;
class Account
{
   private int accountNumber;
   private String holderName;
   private String password;
   private double balance;
   public Account(int accountNumber, String holderName, String password, double balance)
   {
      this.accountNumber = accountNumber;
      this.holderName = holderName;
      this.password = password;
      this.balance = balance;
   }
   public int getAccountNumber()
   {
      return accountNumber;
   }
   public String getHolderName()
   {
      return holderName;
   }
   public String getPassword()
   {
      return password;
   }
   public double getBalance()
   {
      return balance;
```

```
    }
    public void deposit(double amount)
    {
        balance = balance + amount;
    }
    public boolean withdraw(double amount)
    {
        if (amount <= balance)
            {
            balance = balance - amount;
            return true;
        }
        return false;
    }
    public void displayAccount()
    {
        System.out.println("Account Number : " + accountNumber);
        System.out.println("Holder Name    : " + holderName);
        System.out.println("Balance        : " + balance);
    }
}
class Bank
{
    private ArrayList<Account> accounts;
    private int nextAccountNumber;
    public Bank()
    {
        accounts = new ArrayList<>();
        nextAccountNumber = 1001;
    }
    public void createAccount(String holderName, String password, double initialDeposit)
    {
```

```
    if (holderName.trim().isEmpty())
        {
        System.out.println("Holder name cannot be empty.");
        return;
    }
    if (initialDeposit < 0)
        {
        System.out.println("Initial Deposit cannot be negative.");
        return;
    }
    Account account =new Account(nextAccountNumber,holderName,password,initialDeposit);
    accounts.add(account);
    System.out.println("\nAccount Created Successfully.");
    System.out.println("Your Account Number : " + nextAccountNumber);
    nextAccountNumber++;
}
public Account login(String holderName, String password)
{
    for (Account acc : accounts)
        {
        if (acc.getHolderName().equals(holderName)&& acc.getPassword().equals(password))
        {
            return acc;
        }
    }
    return null;
}
private Account findAccount(int accountNumber)
{
    for (Account acc : accounts)
        {
        if (acc.getAccountNumber() == accountNumber)
```

```
        {
            return acc;
        }
    }
    return null;
}
public void deposit(int accountNumber, double amount)
{
    Account acc = findAccount(accountNumber);
    if (acc == null)
        {
        System.out.println("Account Not Found.");
        return;
    }
    if (amount <= 0)
        {
        System.out.println("Invalid Amount.");
        return;
    }
    acc.deposit(amount);
    System.out.println("Deposit Successful.");
    System.out.println("Current Balance : " + acc.getBalance());
}
public void withdraw(int accountNumber, double amount)
{
    Account acc = findAccount(accountNumber);
    if (acc == null)
        {
        System.out.println("Account Not Found.");
        return;
    }
    if (amount <= 0)
```

```
        {
        System.out.println("Invalid Amount.");
        return;
    }
    if (acc.withdraw(amount))
        {
        System.out.println("Withdrawal Successful.");
        System.out.println("Current Balance : " + acc.getBalance());
    }
        else
        {
        System.out.println("Insufficient Balance.");
    }
}
public void transfer(int fromAccNo, int toAccNo, double amount)
{
    if (fromAccNo == toAccNo)
        {
        System.out.println("Cannot transfer to the same account.");
        return;
    }
    Account sender = findAccount(fromAccNo);
    Account receiver = findAccount(toAccNo);
    if (sender == null)
        {
        System.out.println("Source Account Not Found.");
        return;
    }
    if (receiver == null)
        {
        System.out.println("Destination Account Not Found.");
        return;
```

```
    }
    if (amount <= 0)
        {
        System.out.println("Invalid Amount.");
        return;
    }
    if (sender.withdraw(amount))
                {
        receiver.deposit(amount);
        System.out.println("Transfer Successful.");
        System.out.println("Remaining Balance : " + sender.getBalance());
    }
        else
        {
        System.out.println("Insufficient Balance.");
    }
}
public void searchAccount(int accountNumber)
{
    Account acc = findAccount(accountNumber);
    if (acc == null)
    {
        System.out.println("Account Not Found.");
    }
        else
        {
        acc.displayAccount();
    }
}
public void viewBalance(Account acc)
{
    System.out.println("Current Balance : " + acc.getBalance());
```

```
    }
    public void viewMyAccount(Account acc)
    {
        acc.displayAccount();
    }
    public void displayTotalAccounts()
    {
        System.out.println("Total Accounts : " + accounts.size());
    }
    public void displayAllAccounts()
    {
        if (accounts.isEmpty())
            {
                System.out.println("No Accounts Available.");
                return;
        }
        System.out.println("\nALL ACCOUNTS");
        for (Account acc : accounts)
            {
                acc.displayAccount();
                System.out.println();
        }
    }
}
public class Main
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        Bank bank = new Bank();
        while (true)
        {
```

```
System.out.println("\nBANK MANAGEMENT SYSTEM");
System.out.println("1. Create Account");
System.out.println("2. User Login");
System.out.println("3. Admin Login");
System.out.println("4. Exit");
System.out.print("Enter Your Choice : ");
int choice = sc.nextInt();
sc.nextLine();
switch (choice)
    {
   case 1:
      System.out.print("Enter Holder Name : ");
      String name = sc.nextLine();
      System.out.print("Create Password : ");
      String password = sc.nextLine();
      System.out.print("Enter Initial Deposit : ");
      double deposit = sc.nextDouble();
      bank.createAccount(name, password, deposit);
      break;
   case 2:
      System.out.print("Enter Holder Name : ");
      String userName = sc.nextLine();
      System.out.print("Enter Password : ");
      String userPassword = sc.nextLine();
      Account user = bank.login(userName, userPassword);
      if (user == null)
      {
         System.out.println("Invalid Login.");
         break;
      }
      System.out.println("Login Successful.");
      int userChoice;
```

```
do
{
   System.out.println("\nUSER MENU");
   System.out.println("1. Deposit");
   System.out.println("2. Withdraw");
   System.out.println("3. Transfer");
   System.out.println("4. View Balance");
   System.out.println("5. View My Account");
   System.out.println("6. Logout");
   System.out.print("Enter Choice : ");
   userChoice = sc.nextInt();
   switch (userChoice)
                                          {
      case 1:
         System.out.print("Enter Deposit Amount : ");
         double dep = sc.nextDouble();
         bank.deposit(user.getAccountNumber(), dep);
         break;
      case 2:
         System.out.print("Enter Withdrawal Amount : ");
         double wd = sc.nextDouble();
         bank.withdraw(user.getAccountNumber(), wd);
         break;
      case 3:
         System.out.print("Enter Receiver Account Number : ");
         int receiver = sc.nextInt();
         System.out.print("Enter Amount : ");
         double amount = sc.nextDouble();
         bank.transfer(user.getAccountNumber(),receiver,amount);
         break;
      case 4:
         bank.viewBalance(user);
```

```
                    break;
                case 5:
                    bank.viewMyAccount(user);
                    break;
                case 6:
                    System.out.println("Logged Out Successfully.");
                    break;
                default:
                    System.out.println("Invalid Choice.");
            }
        } while (userChoice != 6);
        break;
    case 3:
        System.out.print("Enter Admin Name : ");
        String adminName = sc.nextLine();
        System.out.print("Enter Admin Number : ");
        int adminNo = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Admin Password : ");
        String adminPassword = sc.nextLine();
        if (adminName.equals("admin") && adminNo == 111 &&
adminPassword.equals("admin"))
                                        {
            System.out.println("Admin Login Successful.");
            int adminChoice;
            do
                                                {
                System.out.println("\nADMIN MENU");
                System.out.println("1. Search Account");
                System.out.println("2. View All Accounts");
                System.out.println("3. Total Accounts");
                System.out.println("4. Logout");
                System.out.print("Enter Choice : ");
```

```
            adminChoice = sc.nextInt();
            switch (adminChoice)
                                                    {
                case 1:
                    System.out.print("Enter Account Number : ");
                    int accNo = sc.nextInt();
                    bank.searchAccount(accNo);
                    break;
                case 2:
                    bank.displayAllAccounts();
                    break;
                case 3:
                    bank.displayTotalAccounts();
                    break;
                case 4:
                    System.out.println("Admin Logged Out.");
                    break;
                default:
                    System.out.println("Invalid Choice.");
            }
        } while (adminChoice != 4);
    }
    else
    {
        System.out.println("Invalid Admin Login.");
    }
    break;
case 4:
    System.out.println("Thank You Visit Again!");
    sc.close();
    System.exit(0);
default:
```

```
                System.out.println("Invalid Choice.");
            }
        }
    }
}
```